

AROJ FATIMA

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EDUCATION

National University of Sciences & Technology (NUST)

Sep 2023–Jun 2027

Bachelor of Science in Bioinformatics, Islamabad, Pakistan

CGPA: 3.41/4.00

Relevant Coursework: Discrete Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Organization and Assembly Language, Parallel & Distributed Computing, Linear Algebra, Probability and Statistics, Software Engineering, Fundamental of Computer Programming, Object Oriented Programming

RESEARCH INTERESTS

Formal Verification, Interactive Theorem Proving, Systems Software Verification, C Program Verification, Programming Languages and Formal Semantics, and Verified Compilation.

RESEARCH & PROFESSIONAL EXPERIENCE

Formal Verification Engineer Intern

Jun 2026–Present

Formal Land, Paris, France | Remote

[rocq-of-rust](#)

- Updated Rust-to-Rocq translations as upstream Rust crates and Revm evolved, and developed evaluation methods using OCaml extraction and runtime function resolution.
- Currently develop instruction-level correspondence proofs between Revm and go-ethereum, relating their differently represented stacks, memory, storage, host state, gas, errors, and interpreter behavior.

Student Researcher

Jun 2025–Present

System Analysis and Verification Lab, NUST

[Playground](#)

[Algorithms](#)

[Scheduling](#)

- Built a foundation in interactive theorem proving through Lean 4 game-based exercises and small formalizations. I then moved to Rocq because of its mature ecosystem for verifying programming languages, compilers, and systems.
- Proved correctness properties of BST insertion, insertion sort, merging sorted lists, and MPI block-decomposition formulas, and formalized red-black-tree invariants and repair operations.
- Developed Rocq models and machine-checked proofs for FIFO, SJF, Lottery, MLFQ, Stride, and CFS, including a red-black-tree-based CFS variant. Identified reusable minimum-key, layered-priority, and ticket-interval proof patterns.
- Currently working on the broader formal verification of the $\mu\text{C}/\text{OS-III}$ real-time kernel using Rocq and the Verified Software Toolchain.

Summer Research Intern

Jun–Aug 2025

Deep Learning Laboratory, NCAI, NUST

[Repository](#)

[Certificate](#)

- Training models across decision trees, neural networks, ResNet50, and YOLO on CIFAR and custom datasets
- Reproduced active-learning research for medical-image analysis and compared selection methods using TF-IDF and Sentence-BERT representations.

ACCEPTED REPORT

Arooj Fatima and Saad Saghir. “Towards Machine-Checked Verification of the $\mu\text{C}/\text{OS-III}$ Kernel Using Rocq and the Verified Software Toolchain.” *2026*

Verified with VST four C priority-bitmap functions for a 32-priority $\mu\text{C}/\text{OS-III}$ configuration: initialization, insertion, removal, and highest-ready-priority lookup. Proved that their bit-level behavior satisfies abstract ready-priority specifications.

Accepted for presentation at the **FMCAD 2026 Student Forum.**

[Accepted Reports Listing](#)

[Repository](#)

OPEN-SOURCE CONTRIBUTIONS

Contributor to coqutil and Parseque

Jul 2026–Present

[coqutil](#) and [Parseque](#)

[Rocq](#)

- Authored six merged pull requests adding build documentation, reusable lemmas for lists and optional values, and multiline source-position tests.
- Contributions: [#168](#), [#169](#), [#170](#), [#171](#), [#176](#), [#178](#), and [#22](#).

PROJECTS & SELF-LEARNING

Software Foundations

Dec 2025–Present

Rocq

Private repository

- Working through the Software Foundations in Rocq and completing the chapter exercises. The repository is private because the book asks readers not to post solutions publicly.

Verified Distributed Protein-Network Analysis

May 2026

PySpark, Rocq, and STRING | Part of Parallel & Distributed Computing Course

Repository

- Built a PySpark pipeline that processed 13.7 million protein interactions into 473,860 high-confidence edges over 16,201 proteins.
- Formalized the Large-Star and Small-Star connected-components transformations in Rocq and proved their correctness at a fixpoint.

Concurrent Web-Server Simulation on FreeRTOS

Dec 2025

C, FreeRTOS, RISC-V, and QEMU | Part of Operating Systems Course

Repository

- Built a concurrent web-server simulation demonstrating priority-based scheduling, producer-consumer task queues, and mutex-based synchronization for HTTP requests.

Parallel Algorithm Performance Analysis

Nov 2025

C, OpenMP, Pthreads, and MPI | Part of Operating Systems Course

Repository

- Implemented and benchmarked parallel matrix multiplication and merge sort on datasets ranging from 10 to 10 million elements, comparing the overhead and scaling behavior of three parallel-programming models.

OTHER ACTIVITIES

Curriculum Design and Mentorship Team

Oct 2025–Present

SPARK

- Help plan technical sessions, manage student communication channels, guide students preparing for university.

Executive, Admin Events

Oct 2024–Apr 2025

NUST Entrepreneurs Club

Certificate

- Prepared meeting minutes, event proposals, plans, budgets, and records; coordinated with club leadership; and helped interview and select new members.

Deputy Director, Admin Events

Oct 2024–Apr 2025

NUST IEEE Society

- Supported administrative planning and coordination for student activities and university events.

Organizing Committee Volunteer

Oct 2024

ICoDT2 2024 Conference

Certificate

- Welcomed guests, guided attendees, distributed souvenirs, monitored the venue, and coordinated with other volunteers.

Research Opportunities List Curator

Dec 2025–Mar 2026

Independent GitHub repository

Repository

- Compiled international research programs, deadlines, eligibility requirements, and application links into a public resource for undergraduate students.

AWARD

FMCAD 2026 Student Forum Travel Award for presenting report at TU Graz, Austria

2026

TECHNICAL SKILLS

Verification: Rocq, Lean 4, Verified Software Toolchain

Programming: Rust, C, OCaml, Python

Systems and Tools: Git, Linux, FreeRTOS, MPI, OpenMP, L^AT_EX